

CAR LICENSE PLATE RECOGNITION USING IMAGE PROCESSING

A Computer-Vision Pipeline for the Unique Plate Standards of Northern Cyprus

PURPOSE

Car license-plate recognition using image processing is a technique in which a computer analyses a photograph of a vehicle's license plate and automatically reads the characters on it. This project tailors the technique to the Turkish Republic of Northern Cyprus (KKTC), where off-the-shelf OCR fails because of the distinctive slashed-zero glyph, six co-existing plate layouts and harsh street-photo conditions such as glare and tilt.

METHOD

Each input image is enhanced through five parallel image-processing variants (colour, grayscale, CLAHE contrast, Otsu thresholding and Otsu inverse) and passed to **EasyOCR** for character detection. Detections are merged spatially to reassemble two-line plates and split fragments, and a custom **V/Y disambiguator** inspects glyph geometry to repair the most persistent font confusion. Candidates are validated against **nine regex layouts** using position-aware letter↔digit confusion maps; a deterministic selector ranks survivors by database match, region, correction count and OCR confidence.

PIPELINE

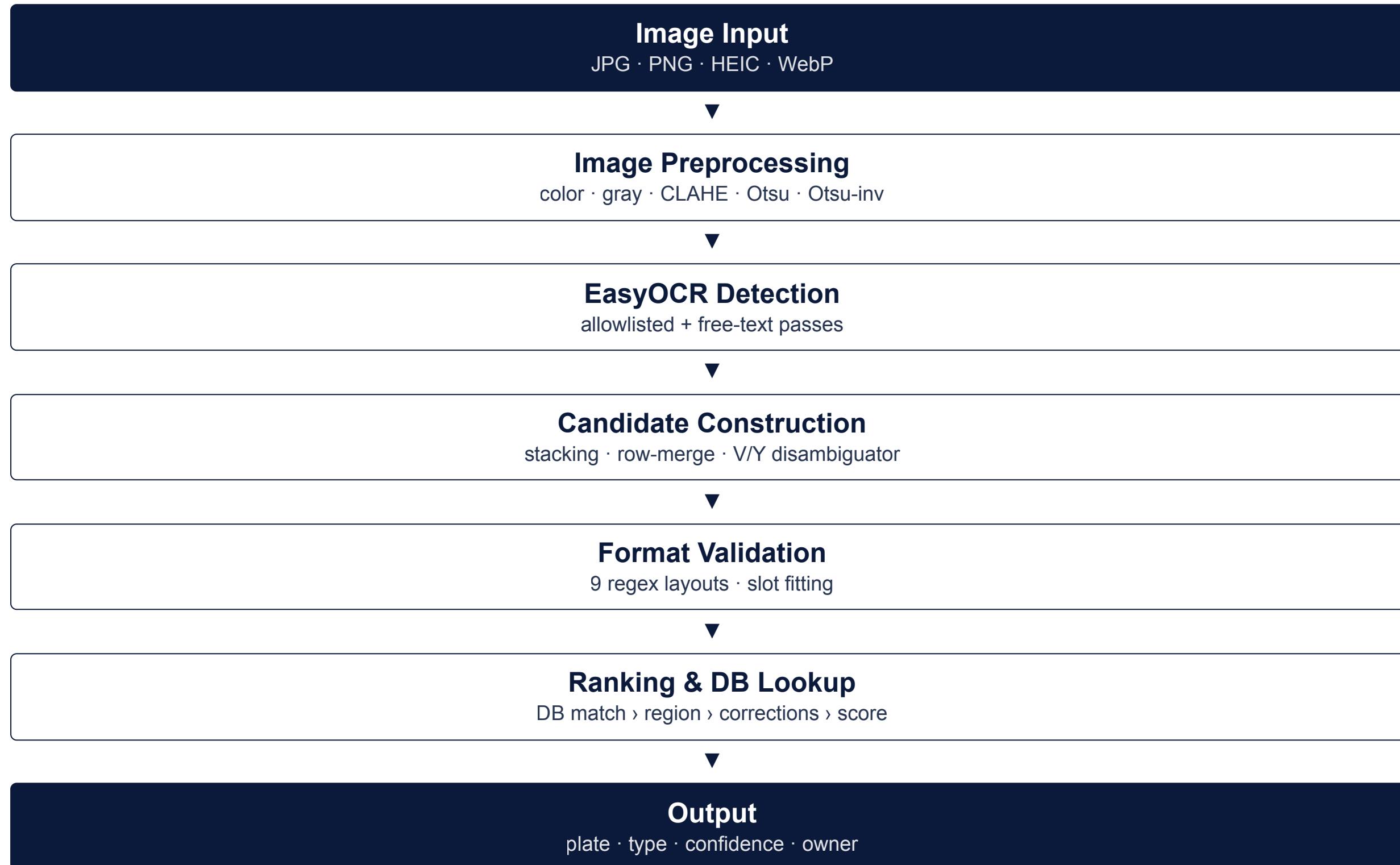


Figure 1. End-to-end recognition pipeline.

SUPPORTED PLATE FORMATS

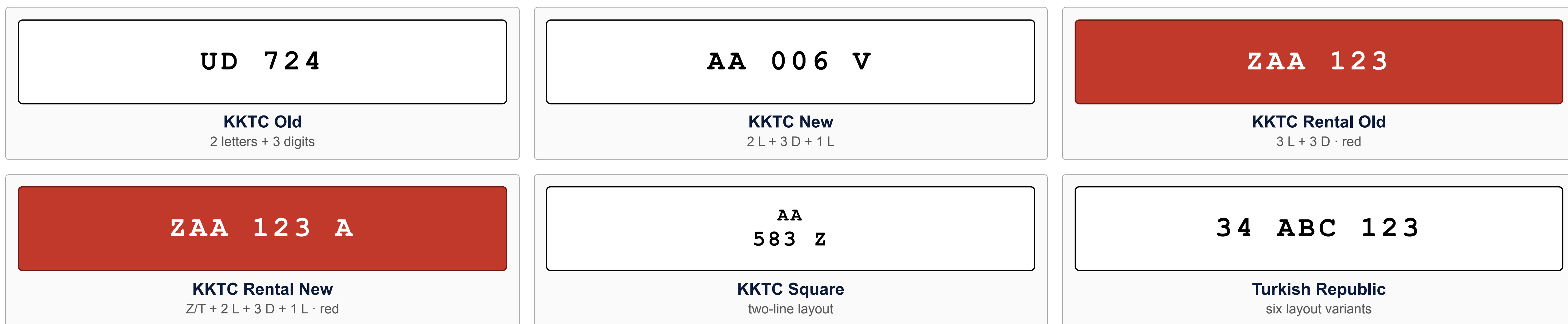


Figure 2. The six plate categories recognised by the system.

RECOGNITION EXAMPLES

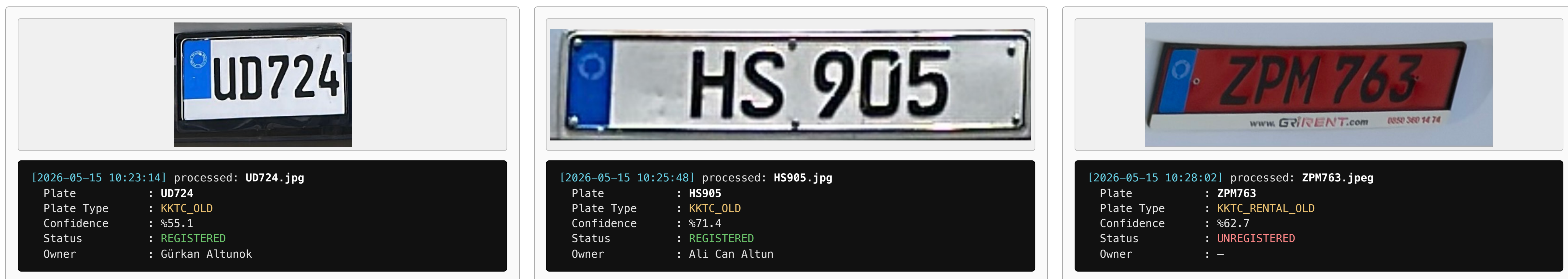


Figure 3. Real KKTC plates and the live terminal output produced by the program.

RESULTS

- All four KKTC subtypes recognised: old, new, rental-old, rental-new.
- Square (two-line) plates reassembled by vertical stacking.
- Turkish-Republic plates accepted across six layout variants.
- Slashed-zero (0/6/8) ambiguity resolved by enumeration vs. the database.
- V/Y font confusion corrected by a geometric stroke-merge test.
- Folder monitoring delivers real-time drop-and-go scanning.

CONCLUSION

The project shows that the practical OCR challenges of KKTC plates can be solved **without retraining a deep model**: standard image-processing operations, a generic OCR engine, layout-aware validation and a database-driven tie-breaker are sufficient. The system runs on CPU, requires no cloud connection after first launch, and can be extended to any new plate format by adding a single regex entry. It illustrates core concepts in computer vision, pattern recognition and automated vehicle identification.